🏆 JeevanSetu Complete End-to-End Pipeline

Phase 1: User Registration & Setup

User Installs App

↓

Signup/Login

↓

Add Emergency Contacts

↓

Grant Permissions

├── GPS

├── Sensors

├── Notifications

├── Phone Calls

└── Location

↓

Profile Created

Output:

- User account
- Emergency contacts
- Permissions enabled

Phase 2: Continuous Monitoring

Nitin (Frontend)

App runs lightweight background monitoring.

Accelerometer

Gyroscope

GPS

Speed

Collected every few seconds.

User Driving

↓

Sensor Monitoring Active

↓

Data Stream Generated

Phase 3: Sensor Data Processing

Harsh (Backend)

Data arrives at backend.

Flutter App

↓

FastAPI

↓

Validation

↓

Preprocessing

Example:


```
{
  "speed": 72,
  "acceleration": 34,
  "gyro": 15,
  "lat": 28.6,
  "lng": 77.2
}
```

Phase 4: Accident Detection AI

Yash + Lavlesh

AI determines if a crash occurred.

Inputs:

Acceleration

Gyroscope

Speed Drop

Orientation Change

Pipeline:

Raw Data

↓

Feature Extraction

↓

Crash Detection Model

↓

Decision

Output:

{
  "accident": true,
  "confidence": 0.94
}

Phase 5: User Verification Layer

Avoid false positives.

Accident Detected

↓

Emergency Popup

↓

"Are You Safe?"

↓

15 Second Countdown

Options:

Case A

I Am Safe

Result:

Cancel Emergency

Resume Monitoring

Case B

No Response

Continue pipeline.

Phase 6: Severity Prediction AI

Yash + Lavlesh

Determine seriousness.

Inputs:

Impact Force

Speed

Acceleration

Orientation Change

Response Delay

Processing:

Feature Extraction

↓

Severity Model

↓

Classification

Output:

```
{
  "severity": "critical",
  "score": 92
}
```

Phase 7: Emergency Orchestration Layer

Harsh

Create emergency incident.

Generate Incident ID

↓

Store Accident Record

↓

Activate Emergency Workflow

Database:

Accidents Table

Stores:

Location

Time

Severity

User

Status

Phase 8: Notification Engine

Abhishek

Send alerts.

Recipients:

Family

Friends

Volunteers

Message:

Possible Accident Detected

Location:

<Google Maps Link>

Severity:

Critical

Using:

Firebase Cloud Messaging

Phase 9: Hospital Recommendation AI

Yash + Lavlesh

Input:

Victim Location

Severity

Hospital Data

Traffic Data

Scoring:

Trauma Capability

ETA

Distance

Availability

Output:

```
{
  "hospital": "AIIMS Trauma Center",
  "eta": "7 min"
}
```

Phase 10: Live Tracking Activation

Nitin + Harsh

Start tracking.

Victim Location

↓

WebSocket

↓

Backend

↓

Dashboard

Updates every few seconds.

Phase 11: Volunteer Network

Abhishek

Search nearby responders.

1 km Radius

↓

3 km Radius

↓

5 km Radius

Notify:

Emergency Nearby

Can You Help?

Volunteer accepts.

Volunteer Assigned

Phase 12: Dashboard Operations

Abhishek

Admin sees:

Accident Location

Severity

Victim Status

Volunteer Status

Recommended Hospital

ETA

Live map updates.

Phase 13: First Aid Assistance

Nitin UI + Future AI

Nearby person opens app.

Need Help?

Choose:

Bleeding

CPR

Fracture

Burn

App gives instructions.

Phase 14: Hospital Transfer

Hospital Selected

↓

Navigation Started

↓

ETA Calculated

↓

Arrival

Phase 15: Incident Closure

Victim Reaches Hospital

↓

Status Updated

↓

Emergency Closed

Stored for analytics.

🧠 AI Sub-Pipeline (Your & Lavlesh's Domain)

Sensor Data

↓

Preprocessing

↓

Feature Extraction

↓

Accident Detection

↓

User Verification

↓

Severity Prediction

↓

Hospital Recommendation

↓

AI Response

Final AI Response:

{
  "incident_id": "INC001",
  "accident": true,
  "severity": "critical",
  "score": 92,
  "hospital": "AIIMS Trauma Center",
  "eta": "7 min"
}

👥 Ownership Summary

- **Yash** → AI Architecture, Detection Logic, Severity Logic, Hospital Ranking Logic
- **Lavlesh** → AI Coding, Model Training, Inference APIs
- **Harsh** → FastAPI Backend, Database, WebSockets, Incident Management
- **Nitin** → Flutter App, Maps, Tracking UI, Emergency Flow
- **Abhishek** → Notifications, Volunteer System, Dashboard, Analytics

This is the complete JeevanSetu pipeline from **"user starts driving" → "accident detected" → "AI decisions" → "rescue coordination" → "hospital arrival"**.